FORMAL METHODS · MATHEMATICAL PHYSICS · AI

Lean and AI for Physics

Formalization, Semantics, and Discovery

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How can physical reasoning become machine-checkable without losing its physical meaning?

Survey · recent formalized physics + AI → **Case study** · special relativity

丘成桐数学科学中心
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Physics Formalization Adds a Modeling Layer

Mathematical formalization

- Formal objects, hypotheses, and theorem statements are made exact.
- The kernel checks whether the conclusion follows from the assumptions.
- Faithfulness to the intended mathematical claim still requires review.

Physics adds a modeling layer

- The target system, regime, and idealizations must be specified.
- The formal objects must be connected to their intended physical interpretation.
- Consequences within the model must be distinguished from claims about nature.

Formalizing physics means formalizing both the deduction and the modeling choices behind it.

From Hilbert’s Sixth Problem to Lean

1900

Hilbert’s Sixth Problem

> “Give a mathematical treatment of the axioms of physics.”
>
> Hilbert’s Sixth Problem · abridged

Core difficulty: choose physically faithful axioms and encode idealizations and limiting relations between theories.
What changes now: formalization makes the chosen assumptions, consequences, and dependencies explicit and machine-checkable.

Modern formalization gives Hilbert’s ambition a machine-checkable medium.

`Hilbert, Mathematical Problems (1902)`

Formalized Physics Is Becoming Practical

Four developments are beginning to reinforce one another.

Lean + Mathlib

Reusable mathematical foundations mean physics projects no longer have to start from scratch.

Physics libraries

Physlib and quantum libraries are organizing domain-specific interfaces.

Research examples

Research-scale work now reaches QFT, quantum information, literature correction, and physical postulate audits, etc.

AI systems

Retrieval, translation, proof search, and candidate generation are reducing the cost of formalization.

> The field is beginning to connect libraries, research examples, benchmarks, and AI workflows into an ecosystem.

What Lean and Mathlib Already Give Us

Kernel-checked Lean excerpt

```
import Mathlib

theorem kineticEnergy_nonneg (m v : ℝ) (hm : 0 ≤ m) :
    0 ≤ (1 / 2 : ℝ) * m * v ^ 2 := by
  positivity
```

What is—and is not—checked

Lean checks a theorem about real numbers.

It does not yet formalize a physical model:
It does not by itself know that `m` is mass, `v` is a Newtonian velocity, or that the result should be an energy.

Mathlib can verify the mathematics of a formula. A physics library must connect that mathematics to a structured physical model.

Theorem Proving in Lean 4 ↗ | Mathlib documentation ↗

Three Layers of Trust in Formalized Physics

FORMAL CLAIM

Deductive correctness

`hm : 0 ≤ m ⊢ 0 ≤ ½mv²`

Does the conclusion follow from the stated formal assumptions?

Directly audited by the Lean kernel

PHYSICAL INTERPRETATION

Semantic fidelity

Do `m`, `v`, and `½mv²` actually represent mass, Newtonian velocity, and kinetic energy in the intended model?

Made inspectable by definitions and domain review

REGIME OF VALIDITY

Scientific adequacy

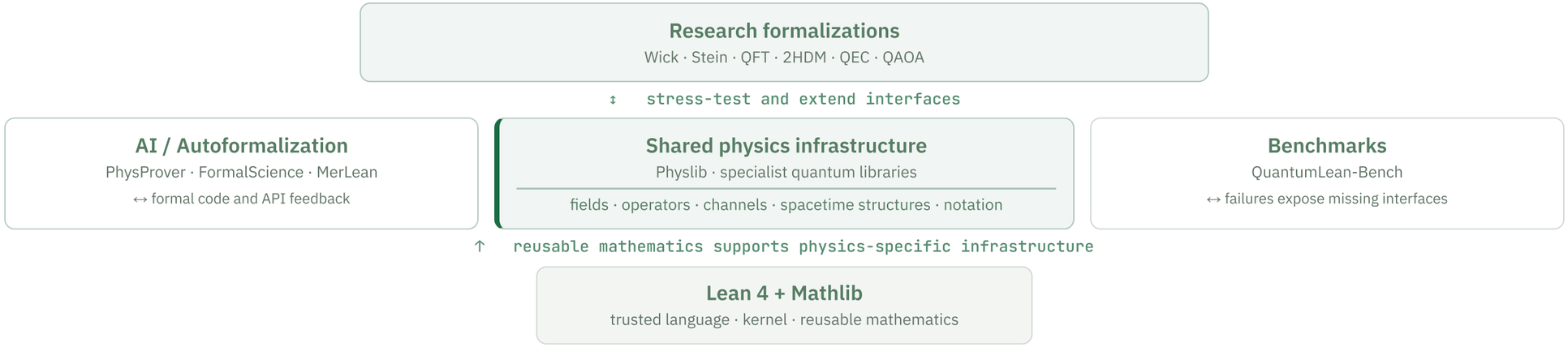
Is the Newtonian kinetic-energy formula appropriate for the physical situation—for example, is `v≪c`?

Still requires physics and experiment

Formalization separates three obligations that informal reasoning often blends: proof, interpretation, and applicability.

From Lean to a Physics Formalization Ecosystem

A mature field needs more than isolated proofs: it needs shared libraries, research-scale case studies, and AI workflows that feed back into one another.



The ecosystem grows through a feedback loop, not a single linear timeline.

Physlib ↗ | FormalScience · arXiv:2604.23002 ↗ | QuantumLean-Bench ↗

How Physlib Became a Shared Library Layer

Zooming into the shared-infrastructure layer

HepLean, 2024
high-energy physics focus

↓ broadened into

PhysLean, 2024–2026
broader physics scope

+

Lean-QuantumInfo
matrix-based quantum-information infrastructure

→ merged into

Physlib, 2026
shared community library layer

Physlib is not the whole ecosystem; it is the community-library node that tries to turn isolated formalizations into reusable physics infrastructure.

Independent projects can use, contribute to, or interact with Physlib without being descendants of it.

Projects need not begin from a complete axiomatization of physics. Many developments can start from a specified mathematical model, as long as its assumptions and intended regime are explicit.

Physlib repository ↗ Physlib website ↗

Physlib Maps Both Infrastructure and Work in Progress

A topic can be represented at different levels of formal maturity.

Selected areas in the repository

- Classical mechanics
- Electromagnetism
- Relativity
- Quantum mechanics
- Quantum information
- Quantum field theory

Different levels of maturity

Reusable API
Definitions and lemmas intended for use across projects.

→

Research-driven module
Substantial formal work organized around a particular result.

→

Explicit future target
A recorded definition, theorem, or API improvement still to be completed.

Repository breadth and uniform formal maturity are different.
A subject area may contain mature APIs, partial interfaces, informal targets, or isolated research modules.

`Physlib source tree ↗`

Example: Energy Conservation for Harmonic Oscillator

In an ideal mass–spring system, kinetic and potential energy are exchanged while the total energy remains constant.

EquationOfMotion
$m\,\ddot{x} + kx = 0$

→

S.energy x_t
$\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$

→

energy_deriv
expands dE/dt

→

force_eq_linear
uses F = -kx

Physlib source excerpt — context omitted

```
lemma energy_conservation_of_equationOfMotion
    (S : HarmonicOscillator)
    (xₜ : Time → EuclideanSpace ℝ (Fin 1))
    (hx : ContDiff ℝ ∞ xₜ)
    (h : S.EquationOfMotion xₜ) :
    ∂ₜ (S.energy xₜ) = 0 := by
  rw [equationOfMotion_iff_newtons_2nd_law _ _ hx] at h
  funext t
  rw [energy_deriv _ _ hx]
  simp [h, force_eq_linear]
```

Trajectory
x_t is a time-dependent point in configuration space.

Smoothness
`hx` supplies the differentiability needed for energy derivatives.

Dynamics
`h` says the trajectory satisfies the equation of motion.
`equationOfMotion_iff_newtons_2nd_law` rewrites it as Newton's second law.

Conservation
The conclusion is `∂ₜ (S.energy xₜ) = 0`.

> A short final proof is possible because the API already connects the model, its dynamics, and its conservation law.

Harmonic-oscillator source ↗

Where Shared Physics Infrastructure Is Still Uneven

Some formal ingredients already have stable reusable interfaces, while others depend more strongly on project-specific modeling choices.

Shared infrastructure beginning to emerge

- Finite-dimensional operator theory
- Quantum channels and entropy
- Tensor and field-theory infrastructure
- Research-driven theorem modules

Projects still build separate physics stacks

- Spacetime, reference frames, and transformations
- Units, dimensions, and conventions
- Preparations, probes, measurements, and physical roles
- Approximations, limits, and regimes of validity

Today, most projects share Mathlib—not yet a common physics layer. Physlib is an emerging integration effort rather than a universal dependency.

`Physlib repository and current structure ↗`

2HDM: Formalization Exposed a Quantifier Gap

PAYOFF 1 · ERROR CORRECTION

This project audits a widely cited 2006 paper on the stability of the general two-Higgs-doublet model. It asks whether the paper's proposed criterion really guarantees one lower bound valid for every allowed field configuration.

QUANTIFIER ORDER CHANGES THE CLAIM

WHAT STABILITY REQUIRES

One global lower bound

$$\exists\ L\ \cdot\ \forall\ x\ \cdot\ V(x) \geq L$$

Choose L once; it works throughout the full field space.

WHAT THE OLD CONDITION GAVE

Direction-dependent lower bounds

$$\forall\ d\ \cdot\ \exists\ L_d\ \cdot\ V \text{ along } d \geq L_d$$

Each direction may choose a different finite lower bound.

WHAT LEAN ADDED Mar 2026

Exposed the quantifier gap
Formalization made the weaker, direction-dependent quantifier structure explicit.

Verified an explicit counterexample
The old directionwise condition can hold while the full potential remains unbounded below.

The 2HDM case begins from a fixed mathematical model, so the main part is theorem auditing.

2HDM paper · arXiv:2603.08139 ↗

Constructive QFT: Auditing a Controlled Construction

PAYOFF 2 · DEPENDENCY CONTROL

Background. Path integrals and perturbative expansions are powerful, but they do not always give a fully constructed mathematical QFT. **Constructive QFT asks whether the theory can be built from controlled mathematical objects.**

WHAT LEAN CHECKS: A DEPENDENCY CHAIN

1 · Controlled input
Euclidean fields:
random generalized fields or distributions

→

2 · Correlation data
Schwinger functions:
Euclidean n-point functions

→

3 · Constructive checks
OS / GJ axioms:
conditions to define a consistent QFT

CHECKED

Current formal result

A free bosonic QFT in four-dimensional Euclidean spacetime satisfies the Glimm–Jaffe axioms, with the dependency chain checked in Lean and Mathlib.

NEXT STAGE — NOT FORMALIZED HERE

Lorentzian reconstruction

From Euclidean correlation functions to a Lorentzian QFT

Point: Lean makes the complete Euclidean construction machine-checkable and makes its exact scope explicit.

Formalization of QFT · arXiv:2603.15770 ↗

Seiberg–Witten: What Lean Proves—and What It Assumes

PAYOFF 3 · TRUST LOCALIZATION

Seiberg-Witten theory gives exact low-energy predictions for a strongly coupled four-dimensional supersymmetric gauge theory. Because some of its physical inputs do not yet follow from a rigorous first-principles construction, **the author Michael Douglas isolates them as explicit physical hypotheses H0-H7.**

Conditional theorem checked: From H0–H7 and classical mathematical inputs, Lean proves that the genus-one SU(2) effective coupling exists and is unique up to electric–magnetic duality.

Physical hypotheses
H0–H7 enter as explicit theorem parameters and remain trusted physical inputs

→

Lean derivation
Every step from the stated hypotheses to the headline result is machine-checked

→

Assumption audit
Theorem types track physical hypotheses; `#print axioms` tracks residual mathematics

Lean does not prove H0–H7. It proves exactly what follows from them and makes the remaining trust finite, named, and inspectable.

Axioms for physical reasoning · arXiv:2607.06379 ↗ | Lean repository ↗

Wick's Theorem: Turning a Standard Identity into a Reusable API

PAYOFF 4 · REUSABLE INFRASTRUCTURE

The physics

The authors formalize Wick's theorem because it is the combinatorial engine behind perturbative QFT: **time-ordered products are expanded into terms indexed by contraction patterns.**

$$T(\varphi_1 \cdots \varphi_n) = \Sigma_{\text{contraction patterns } \Lambda} \text{ wick term}(\Lambda)$$

Scope: free-field operator combinatorics—not a construction of a complete interacting quantum field theory.

The reusable Lean representation

Lean-style pseudocode

```
theorem wicks_theorem :
  (φs : List .FieldOp) →
  (ofFieldOpList φs) =
  ∑ (φsΛ : WickContraction φs.length),
    φsΛ.wickTerm
```

FieldOp · typed field operator
WickContraction · disjoint pairings of list positions
wickTerm · propagators, remaining normal-ordered fields, and signs

> **Lean's contribution:** contraction patterns and Wick terms become a reusable API for later Feynman-diagram and perturbative-QFT developments.

Physlib Wick theorem ↗ | Digitalizing Wick's theorem · arXiv:2505.07939 ↗

Quantum Information as an End-to-End Formalization Testbed

Why focus here? Quantum information already shows a clear path from reusable interfaces to research theorems, scalable verification, and formal benchmarks.

Executable, notation-facing

- Gates, Pauli matrices, and bra–ket notation
- Small examples that remain recognizable to physics students
- Concrete finite-matrix goals can be executed and checked

Representative: LeanQuantum

Abstract, operator-theoretic

- States, channels, partial trace, and entropy
- Choi, Kraus, and Stinespring representations
- Reusable infrastructure for general research statements

Representatives: Physlib QuantumInfo, Lean-Quantum

readable notation → reusable abstractions → deep theorems / scalable certificates → benchmarks

LeanQuantum repository ↗ | Physlib QuantumInfo ↗ | Lean-Quantum · arXiv:2607.05492 ↗

LeanQuantum Provides a Readable Entry Point

READABLE ENTRY · INDEPENDENT PACKAGE

What LeanQuantum provides

- Textbook-facing gates, states, and bra–ket notation
- Pauli operators and stabilizer-code infrastructure
- Automation for concrete finite-matrix goals

Official LeanQuantum usage example

```
import Quantumlib

example : hadamard * hadamard = 1 := by
  solve_matrix [hadamard]

example : σx * |0⟩ = |1⟩ := by
  solve_matrix [σx]
```

Separate package · not part of Physlib

This example shows its matrix-facing entry layer: the statement remains readable while Lean checks the calculation.

`solve_matrix` is one tactic LeanQuantum; it checks a stated matrix goal but does not choose the physical model.

LeanQuantum repository ↗

Beyond Explicit Matrices: Analytic and Operational Layers

Research-level quantum information also needs abstract mathematical objects and precise operational tasks.

Analytic / operator layer

Representative project: Lean-Quantum

How do distinguishability and entropy change when a state passes through a quantum channel?

- States, channels, and partial trace
- Entropy and quantum divergences
- Choi, Kraus, and Stinespring representations
- Operator and trace inequalities

Provides the mathematical machinery for general quantum-information theorems.

Operational / coding layer

Representative project: Lean-QIT

What information-processing task is being performed, and what rate or error is achievable?

- Source and channel codes
- Error criteria and one-shot quantities
- Rates, capacities, achievability, and converse
- Asymptotic coding theorems

Makes compression and communication tasks formally precise.

Naming note: `LeanQuantum` ≠ `Lean-Quantum` ≠ `Lean-QuantumInfo`.

`Lean-Quantum · arXiv:2607.05492 ↗` `Lean-QIT · arXiv:2607.09632 ↗`

Two Forms of Research Scale in Quantum Formalization

Two bottlenecks: deep theorem dependencies and large checkable search results.

Dependency depth · Generalized quantum Stein's lemma

Physical question

How do we distinguish a state ρ from an alternative set $\mathcal{S}$?

many copies · Type-I error ≤ ε · minimize Type-II error

Mathematical theorem

optimal distinguishability rate = regularized relative entropy

```
limn→∞-(1/n) log βn*(ε) = Dreg(ρ // 𝒮)
```

Operational meaning: the optimal discrimination error decays exponentially at a rate given by the regularized relative entropy.

Formalization burden

measurements → optimization → operator inequalities → asymptotic limit

Lean contribution · Lean-QuantumInfo → Physlib: makes the full dependency chain explicit and machine-checkable.

Verification scale · QEC distance certificates

Physical question

How can we protect logical quantum information against errors?

```
α|0⟩ + β|1⟩ → α|000⟩ + β|111⟩
```

single-X error → stabilizer syndrome → error location

modeled single-X-error set

LeanQuantum excerpt

```
example : BitFlip.code.corrects
    {[P| IIX ], [P| IXI ], [P| XII ]} :=
  BitFlip.corrects_single_qubit_errors
```

Research target

certify the stabilizer-code distance

minimum number of qubits affected by an undetectable logical error

Formalization burden

distance condition → Boolean constraints (SAT) → checked certificate

search grows combinatorially with the number of qubits

Lean contribution · Lean-QEC: verifies the SAT encoding and checks the resulting certificate.

Quantum Stein's lemma · arXiv:2510.08672 ↗ | LeanQuantum QEC examples ↗ | Lean-QEC · arXiv:2605.16523 ↗

QuantumLean-Bench: From Quantum Reasoning to Lean-Checkable Output

Each problem is evaluated in two matched forms: a natural-language solution and an end-to-end Lean formalization.

Scope
931 problems

chemistry · computing · information theory · physics · quantum machine learning
numerical · algorithmic · theorem · computational tasks

INFORMAL EVALUATION

Natural-language solution

```
quantum problem → natural-language solution
```

Tests: does the model solve the quantum problem correctly?

FORMAL EVALUATION

End-to-end formalization

```
same problem + Lean context → theorem statement + proof → Lean typecheck
```

Tests: can the model formulate and prove a faithful Lean version of the problem?

Baseline gap · GPT-5.4 manual evaluation **Informal:** 896 / 931 top score — **96.2%** · **Formal:** 336 / 931 top score — **36.1%**
Both counts are based on manual evaluation of answer quality and faithfulness, not Lean typecheck success.

QuantumLean-Bench ↗

Three AI Roles in the Formalization Pipeline

Current systems intervene at different stages of that pipeline, while still relying on a human-reviewed scientific target.

PhysProver · Jan 2026

Automates
proof generation in an existing Lean environment

Still requires

a trusted statement, library interfaces, and environment design

`fixed theorem + environment → domain-trained prover → checked proof`

MerLean · Feb 2026

Automates
drafting Lean declarations from LaTeX papers

Still requires

review of newly introduced definitions and axioms

`LaTeX paper → Lean declarations → LaTeX back-translation for review`

FormalScience · Apr 2026

Automates
compiler-guided drafting and refinement

Still requires

domain-expert review of semantic alignment

`informal solution ↔ generated Lean ↔ compiler feedback ↔ human review`

These systems automate different stages. They prepare the setting in which AI-assisted discovery can occur, but they do not remove the need to define the scientific problem.

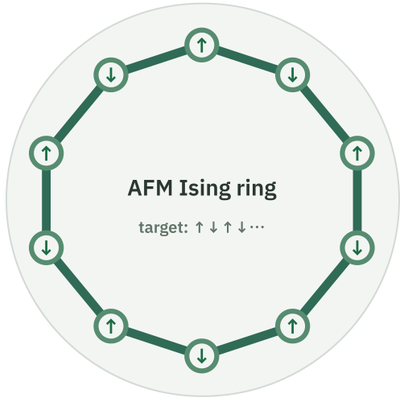
PhysProver · arXiv:2601.15737 ↗ MerLean · arXiv:2602.16554 ↗ FormalScience · arXiv:2604.23002 ↗

QAOA: AI Discovery Within a Fixed Formal Problem

Quantum Approximate Optimization Algorithm on a 1D antiferromagnetic Ising ring with periodic boundary conditions.

PHYSICAL PROBLEM

How closely can a depth-*p* QAOA circuit approach the optimal energy of a 1D antiferromagnetic Ising ring?



Human-specified open conjecture

For depth-*p* QAOA on the antiferromagnetic Ising ring,

$$r_p^* = (2p + 1) / (2p + 2)$$

p: circuit depth · r_p^*: best achievable objective-value ratio at depth *p*

The model, theorem statement, and conjectured optimum were already fixed.

Human-built Ising/QAOA formalization + fixed open theorem

↓

AI-discovered proof route

`free fermions → SU(2) pseudospins → quantum signal processing`

↓

Lean checked the route proves the conjecture.

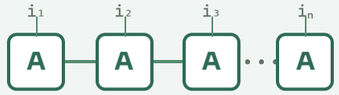
The scientific target was human-defined; AI discovered the proof route; Lean verified it.

QAOA conjecture proof · arXiv:2606.29687 ↗

From Proof Automation to Research-Scale Formalization

TARGET: MATRIX-PRODUCT STATES · FT-MPS

When do different local tensors encode the same global physics?



$$|\psi_N(A)\rangle = \Sigma_{i_1,\ldots,i_n} \operatorname{tr}(A^{i_1} \cdots A^{i_n}) \, |i_1 \cdots i_n\rangle$$

AFTER CANONICAL FORM

Same physical state family ⇔ virtual-bond gauge equivalence

$$\forall\, N \geq 1,\quad |\psi_N(A)\rangle = |\psi_N(B)\rangle \;\Leftrightarrow\; \exists\, X \in GL(D),\quad B^i = X\, A^i\, X^{-1}$$

⇐ immediate from trace invariance · ⇒ the hard FT-MPS direction

AGENT-DRIVEN DEVELOPMENT

A research theorem becomes a coordinated formalization project

primary literature → **mathematical blueprint** → specialized agents → Lean-checked development

human audit: literature ↔ blueprint ↔ Lean statement

≈62k	0	6
core Lean lines	**sorry** in the core proof chain	human review rounds

AI-found nonstandard route for a key special case:
matrix-algebra automorphism → Skolem–Noether → gauge conjugation.

TNLean is the project's own ≈227k-line tensor-network and quantum-information library.

A KERNEL-CHECKED DEVELOPMENT CAN STILL FORMALIZE THE WRONG THEOREM

Unintended stronger assumption. A doubly-stochastic gauge made the proof easier but restricted the theorem to a much smaller class of MPS tensors.

Wrong notion of equivalence. An early version replaced equality at every finite chain length by an asymptotic comparison and lost essential finite-N information.

AI can scale proof production and library building. At research scale, specifying and auditing mathematical intent becomes more important.

Multi-agent autoformalization of tensor networks · arXiv:2607.07857 ↗

TNLean repository ↗

From Fixed Formal Problems to Physical Model Design

QAOA and MPS both start from fixed problems. But physics formalization may have to start one layer earlier: deciding what the formal target should mean.

AFTER THE PROBLEM IS FIXED

Search and verify inside a formal environment

- definitions and theorem statement are already specified;
- the library determines the available interfaces;
- AI searches for proof routes or constructions;
- Lean checks whether the conclusion follows.

← physics often starts earlier

BEFORE THE PROBLEM CAN BE FIXED

Choose the physical objects and interfaces

- what counts as a frame or an inertial frame;
- what generates a field and what merely probes it;
- how representations are compared across frames;
- which regime, assumptions, and operational meanings are intended.

Common upstream failures: hypothesis creep · definition drift · model/interface drift

Recall the Seiberg-Witten project turns physical lore into explicit hypotheses. In our special-relativity work, the question starts even earlier: what objects and interfaces are needed before the theorem can be stated?

Special Relativity and Its Most Famous Postulate

Special relativity resolves an incompatibility between Galilean kinematics and Maxwell electrodynamics.

GALILEAN FRAMEWORK

Relativity with classical kinematics

Inertial frames obey Galilean velocity addition:

$$u' = u - v$$

Newtonian mechanics retains the same form under this rule.

MAXWELL ELECTRODYNAMICS

A distinguished wave speed

Maxwell's vacuum equations admit electromagnetic waves with a characteristic speed c.

CHARACTERISTIC WAVE SPEED

$$c = 1 / \sqrt{\mu_0 \varepsilon_0}$$

Galilean transformations do not preserve the form of Maxwell's equations.

EINSTEIN'S RESOLUTION

Light-speed invariance

Einstein retained the relativity principle and postulated the same light speed in every inertial frame.

$$c' = c$$

while time, simultaneity, and spatial distances become frame-dependent.

Physical program proposed by Congjun Wu · Westlake University

Can light-speed invariance be derived from weaker physical assumptions, rather than taken as primitive?

Even this familiar question has surprisingly subtle formal foundations.

Many examples above share Mathlib but not a common physics-model layer; this foundational case study shows why such a layer is difficult to design.

Before Light-Speed Invariance: First Prove a Speed Bound

Congjun Wu's aim is to derive light-speed invariance from more elementary physical assumptions.

First formal target

$$v_{th}(R) \le c$$

- R: distinguished Maxwell frame
- $v_{th}(R)$: mechanically admissible speed threshold
- c: electromagnetic characteristic speed in the chosen model

> At this stage, c is still an electromagnetic constant in the setup.
> It is not yet proved to be the speed of light in every inertial frame.

Two Approaches

EINSTEIN-STYLE ROUTE 1905
light-speed invariance as a postulate

VS

ALTERNATIVE ROUTE BY WU
Mechanical relativity + Lorentz force + one-frame steady Maxwell calculation → $v_{th}(R) \le c$

Current formal target, not the full light-speed-invariance theorem.

Our current development formulates the interfaces and theorem statement needed for this speed-bound theorem.

Congjun Wu derivation note (PDF) ↗

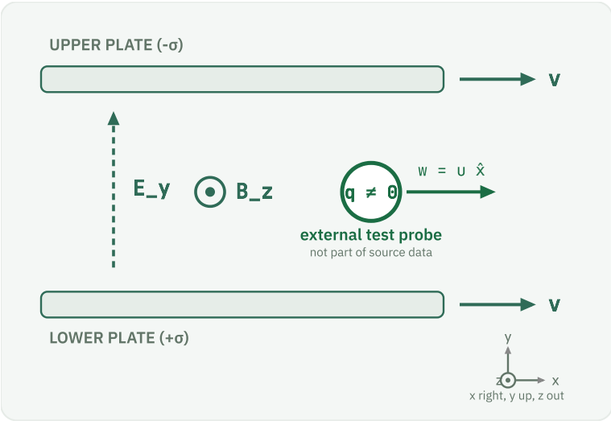
The Thought Experiment: A Speed Limit from Charged Plates

1 **Moving charged plates produce fields.**
In the lab frame, the moving plates give an electric field E_y and a magnetic field B_z.

2 **Choose the force-free drift probe.**
Lorentz force for probe: `F = q(E + w×B/c)`. Choose the probe velocity $w = u\,\hat{x}$ with `u = c²/v`; then F_R `= 0`.

3 **Assume $v > c$ is physical.**
Then $u = c^2/v < v$. Downward closure makes this drift speed physical.

4 **Contradiction in plates' co-moving frame.**
Symmetry gives $B' = 0$; the non-vanishing electric response $E' \neq 0$ with $q \neq 0$ would give $F' \neq 0$. Conflicts with mechanical relativity.



Paper Proof vs. Formal Proof

Formalization exposes the interfaces that an informal derivation silently moves across.

PAPER PHRASE	LEAN NEEDS
"a reference frame"	Coordinate map from events to time + space; inertiality is extra structure
"the plates move with velocity v"	Extended source representation in a frame, not a particle worldline
"the same plates in their rest frame"	Co-moving frame for an extended source system
"put a test charge"	Separate localized probe with $q \neq 0$; it reads the field but is not part of the source data
"a physical speed v"	Speed-domain data: $v \in P_R$, with downward closure $u \leq v \Rightarrow u \in P_R$
"compare fields in two frames"	Field-response determination + local zero preservation

The next slides unpack three separations: frame vs. inertial frame, source vs. probe, and field values vs. field-response comparison.

Frames Are Not Just Coordinates

A frame starts as coordinates on events; a physical system appears through its frame representation.

Minimal starting point

```
/--
An assignment which assigns each physical event to a
spacetime coordinate; distinct events cannot be
assigned the same coordinate.
-/
structure ReferenceFrame (Event : Type _) where
  coord : Event → Coord4
  sep   : Function.Injective coord
```

No inertial mechanics is built into this structure.

Structure added later

1. **InertialFrame**
   force-free motion ⇔ uniform straight-line motion
2. **Representation in a Frame**
   we intentionally distinguish the frame from how a physical system is represented in that frame
3. **CoMovingFrame**
   an inertial frame where the representation is at rest or static

The charged-plate proof uses both layers: inertial frames control force-free motion, while frame representations control how probes and sources are compared.

Physical Systems: Source/Probe Separation

DEFINITION **Physical system.** A physical system is a frame-independent object or source that can be represented in reference frames. We separate two frame representations with different roles:

Worldline Representation

- Used for a **localized system**, such as a test particle or small probe.
- Represented in a frame by a worldline or motion; with charge, it reads the field by Lorentz force.

↓

$F = q(E + w \times B/c)$

← FIELD FLOW

Source Representation

- Used for an **extended charge-current source**, such as the plates.
- Represented in a frame by admissible source data (ρ, j).

↓

Field (E, B)

This separation matters because otherwise the object used to test the field would also change the source that produces the field.

Maxwell solves the field generated by the plate source representation. The Lorentz force is then evaluated on a separate charged worldline representation.

Symmetry Fixes the Field Shape

We can use symmetry to kill most field components: **if the source is invariant under a symmetry, then the generated field must transform compatibly.**

What kills what?

translations along plates
kill dependence on x and z.

z-reflection
kills E_z, B_x, and B_y.

$R_y(\pi)T$
kills E_x.

Component parity table

	B_x	B_y	B_z	E_x	E_y	E_z
σ_{xy}	−	−	+	+	+	−
$R_y(\pi)T$	+	−	+	−	+	−
survives	✗	✗	✓	✗	✓	✗

The listed symmetries preserve the moving-plate source. A **−** therefore forces that component to vanish. Here E is polar, while B is axial.

Symmetry is a natural interface which turns source geometry into reusable field-shape lemmas: $E = E_y\,\hat{y}$, $B = B_z\,\hat{z}$. Maxwell laws later fix the coefficients.

When Formalization Catches a Hidden Assumption

Paper shortcut by Wu: in the plates' co-moving frame: $B' = 0$ by time reversal, and $E' \neq 0$.

ACCEPTED BY SYMMETRY

$B' = 0$

In the plates' co-moving frame, the source is static. Since magnetic field is odd under time reversal, it vanishes.

NOT AUTOMATIC

Why is $E' \neq 0$?

We cannot simply assume Maxwell equations again in this frame. The explicit Maxwell was only assumed in the lab frame R.

$$\text{assume } (E', B') = (0,0) \quad \xRightarrow{\text{linearity: zero maps to zero}} \quad \textbf{contradicts nonzero response in } R$$

Formalization forces every hidden physical step to say which assumption justifies it. This leads to axiom audit.

What Makes a Physical Axiom Good?

In the spirit of **Hilbert's sixth problem**, when Lean exposes a missing step in which we need a new assumption, we ask is it a general principle or just a patch?

LOCAL PATCH

assume the missing step directly

- ✕ high target overlap
- ✕ no explanation
- ✕ hard to reuse

EXAMPLES

`E = Ey ŷ, B = Bz ẑ`

`E′ ≠ 0`

`proof gap`

Lean

`axiom audit`

Lean turns a gap into an audit question

STRUCTURAL GENERATOR

find the principle that generates the missing step

- ✔ low target overlap
- ✔ explanatory
- ✔ generates reusable lemmas

`structural principle`

↓

`local seam 1` `local seam 2` `local seam 3`

EXAMPLES

`source symmetry + parity`

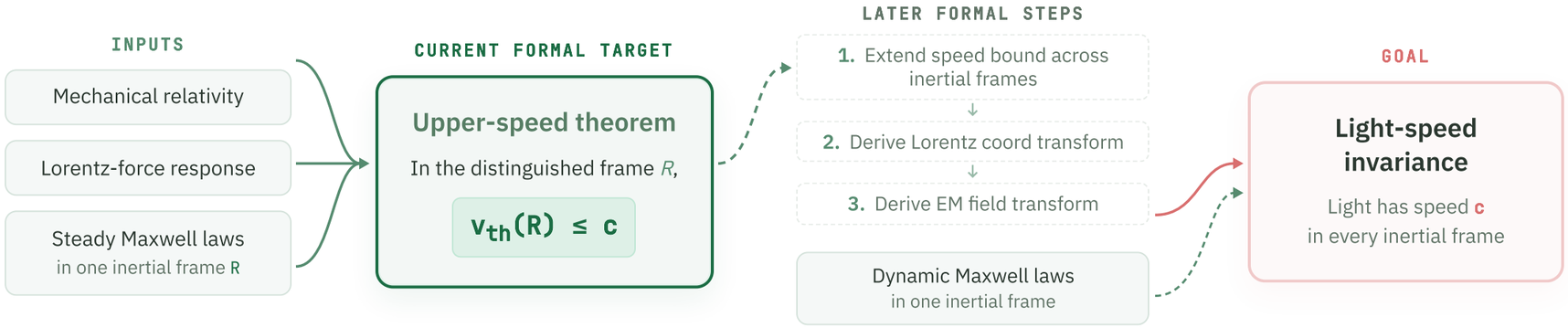
`field-response comparison`

> **Formalization not only asks what assumption is missing, it also helps judge the quality of the assumption. A good axiom often becomes part of the interface.**

This also echoes the Seiberg–Witten formalization, where several apparently separate physical assumptions are ultimately viewed as specializations of a common structural principle.

Where the Upper-Speed Theorem Fits

The charged-plate program targets a speed bound first. The larger route continues toward light-speed invariance.



> **Formalization turns Wu's paper derivation into an auditable route: which assumptions and interfaces are given, what remains to be proved, and which steps are still informal.**

Open Problems: From Verified Proofs to Reusable Physics

The central challenge is to represent, share, and preserve physical meaning in formal systems.

1 · Represent physical meaning

MODEL-ROLE INTERFACES

frames · sources · probes · preparations · measurements · regimes

Formal objects must record not only their mathematical type, but also the physical role they play.

2 · Share physical meaning

INTEROPERABILITY BETWEEN PHYSICS STACKS

Mathlib is shared; physics APIs remain fragmented.

Independent projects need common interfaces for states, fields, channels, spacetime, conventions, and validity regimes.

3 · Preserve physical meaning

SEMANTIC AUDIT FOR AI SCALING

statement drift · definition drift · wrong equivalence · provenance · scope

A file can compile while the theorem, model, assumptions, or notion of equivalence has quietly changed.

The next stage is not only larger formalizations, but reusable physics interfaces that preserve intended meaning across projects and AI workflows.

Thank You

Lean verifies what follows from a formal model.

The deeper value of formalization in physics is to make the model, its assumptions, and its remaining gaps inspectable.

> **Isolated Formalizations → Shared Infrastructure → AI at Research Scale → Auditable Formal Science : Formalized Physics Is Entering a New Phase**

Relativity Formalization ONGOING

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